

Quantum Codes for Private Distributed Matrix Multiplication for the Low Privacy Regime

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Abstract—We consider the problem of private distributed matrix multiplication (PDMM) in the quantum setting. We focus specifically on the case where the privacy condition is low compared to the matrix parameters. In this setting, none of the classical PDMM codes can be extended to their quantum counterparts. Thus, we develop codes specifically for the quantum PDMM problem. We provide an achievable scheme to show how the encoding and decoding can be done in the quantum setting. In addition, we provide several numerical results showing the varying levels of gain for a wide range of parameters. Although, in general, the gain for low privacy regime is expected to be very low due to the entanglement only between the transmitters, we are able to achieve gains that go up to 1.5 times the classical gain in some cases. This result shows that, with careful code design, there is still potential to make good use of the available quantum resources in privacy applications.

I. INTRODUCTION

It is widely known that matrix multiplication (MM) is a computationally intensive task, especially in large dimensions. Since the early days of machine learning, MM has played a crucial role in the learning algorithms. Most recently, the rapid growth of artificial intelligence, along with its huge breadth of applications and the abundance of highly complex data, have given rise to new challenges. The main challenge is the surge in the data dimensionality and the required computational power to apply multiple complex operations over personal edge devices. It is becoming less and less realistic to expect a personal device to handle these operations. Thus, an alternative approach has emerged which is to outsource these operations to high-powered computing entities. At the same time, the user, at the edge device, wants to keep their data private. In order to accommodate these contradicting constraints, private distributed matrix multiplication (PDMM) was introduced [1].

There are many variants and settings for the PDMM with several accompanied interesting code constructions. In general, PDMM defines a setting, where a user has two matrices A and B partitioned into K and L sub-blocks, respectively, and would like to compute the multiplication AB , while possessing only limited computational resources. To resolve this issue, the user sends these matrices, in an encoded form, to N servers to compute the required multiplication. The user wants the matrices to remain completely hidden from the servers. The user also knows that any $T < N$ out of N servers could be honest but curious, thus they will perform the required scheme but could exchange information about the received encoded

matrices, i.e., they are colluding or communicating. Thus, the encoded transmissions must be private from any T colluding servers. Once the user receives the answers from the N servers, the required multiplication must be decodable. See [2]–[16] for many interesting variants of PDMM. The main difference between classical and quantum PDMM is that in the quantum PDMM, the servers respond back to the user over separate quantum channels and the servers share an entangled state.

In this paper, we focus on outer product partitioning (OPP), where A is sliced *horizontally* into K sub-blocks and B is sliced *vertically* into L sub-blocks. In this paper, we specifically focus on the low privacy regime, i.e., when $K \geq L \geq T^1$. Two main issues need to be addressed prior to delving into the low privacy regime considered here. The first is the extension of the original classical codes to their corresponding quantum counterparts, and the second is the high privacy regime codes, i.e., when $T \geq K \geq L$ and $K \geq T \geq L$. To address the first issue, we developed previously in [17], [18] a feasibility condition such that when satisfied for any arbitrary PDMM OPP classical code, it can be extended to a quantum code, thus achieving super-dense coding gain, i.e., the highest possible gain in quantum communication. However, it was also shown that to satisfy this condition, the privacy parameter needs to be in the range of $T = \mathcal{O}(KL)$. Thus, special quantum PDMM codes have to be designed for the complementary range, i.e., $T = o(KL)$. To deal with the design problem in two distinct regimes, the high privacy regime and the low privacy regime. For the high privacy regime, which is the second issue, we designed specific codes in prior work in [18].

The techniques to generate codes for the low privacy regime are different than the ones for the high privacy regime and are more involved. The main issue when dealing with the low privacy regime is that, if not designed carefully, quantum resources may seem useless, i.e., it may appear at first that there may be no gain in using quantum resources. This subtlety stems from the ability to manipulate the privacy subspace compared to the information subspace when the servers are the only entangled entities.

In this paper, we focus on this low privacy regime. We design codes that work for the quantum setting for arbitrary $K \geq L \geq T$. In order to simplify the presentation here, we consider the case when $K = L \geq T$, in which case, it is

¹In this work, we assume without loss of generality that $K \geq L$ [16].

easier to follow the intuition of the chosen exponents and the scheme. We compare these results with the highest possible rate in the classical setting. We obtain a varying gain, with a maximum of 1.5 times rate gain. Although this is not as high as the 2 times rate gain achieved in the high privacy regime, this is to be expected in the low privacy regime.

Due to space limitations, this paper presents the main results together with one representative achievable scheme; the necessary preliminaries, detailed derivations, and full proofs are deferred to [18].

II. PROBLEM FORMULATION

We now formulate the PDMM problem in the classical and quantum settings, respectively. In the classical setting, a user possessing two matrices A and B , with constrained local computational resources, seeks to compute their product. These matrices are divided into K and L sub-blocks, respectively, as follows

$$A = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_K \end{bmatrix}, \quad B = [B_1, B_2, \dots, B_L], \quad (1)$$

where all $A_i B_j$ are compatible for computing the outer product, i.e.,

$$AB = \begin{bmatrix} A_1 B_1 & A_1 B_2 & \dots & A_1 B_L \\ A_2 B_1 & A_2 B_2 & \dots & A_2 B_L \\ \vdots & \vdots & \ddots & \vdots \\ A_K B_1 & A_K B_2 & \dots & A_K B_L \end{bmatrix}. \quad (2)$$

Given K and L , the user distributes the encoded versions of the matrices to N servers, such that these matrices are secure, i.e., private, against any T colluding servers, i.e.,

$$I(f_{\mathcal{T}_1}, g_{\mathcal{T}_2}; A, B) = 0, \quad (3)$$

where f_i and g_i are the information sent by the user to the i th server about A and B , respectively.² In addition, $\mathcal{T}_i \subset [N]$, with $|\mathcal{T}_i| \leq T$, $i \in [2]$. It is important to note that T collusion here signifies that any T servers can share the received information from the user.

In order to ensure privacy for each of these matrices, a subspace with dimension at least T has to be designed to protect the transmitted matrices. This can potentially result in a noise space with high dimensionality. Without careful design, the resulting code may suffer rate inefficiency. To address this, we adopt polynomial codes: Polynomial codes are considered as a special case of linear codes due to their nice structure that can be utilized in the analysis.

A polynomial code for product computation is defined by two polynomials, $f: \mathbb{F}_q \rightarrow \mathbb{F}_q^{s \times \kappa}$ and $g: \mathbb{F}_q \rightarrow \mathbb{F}_q^{\kappa \times \ell}$, and the goal is to compute $h = fg$ using the N servers by sending distinct evaluations of each of these polynomials to the servers, i.e., $f_i = f(x_i)$ and $g_i = g(x_i)$ are sent to the i th server.

²It is assumed that A and B are generated independently and uniform at random.

Classically, once the user receives the responses, $h_{[N]}$, the required multiplication can be reconstructed, i.e.,

$$H(AB|h_{[N]}) = 0, \quad (4)$$

where h_i is the multiplication done at the i th server. In addition, the rate can be defined as

$$R_C(K, L, T) = \frac{KL}{N}. \quad (5)$$

Remark 1 *It is important to note that, since only one polynomial, f , is used to encode the sub-blocks of A_i , sub-blocks have to be of equal dimensions, or zero padding must be used. The same argument holds for the sub-blocks of B .*

In the quantum setting, the servers share an entangled state, represented by the density matrix ρ^{init} , in the quantum system $Q^{\text{init}} = Q_1^{\text{init}} \cdot Q_2^{\text{init}} \cdot \dots \cdot Q_N^{\text{init}}$, where the Q_n^{init} subsystem is at a state represented by the density matrix $\rho_n^{\text{init}} = \text{tr}_{j \in [N] \setminus \{n\}}(\rho^{\text{init}})$, where $\text{tr}_m(\cdot)$ is the partial trace operator in the finite Hilbert space. The user sends the sub-blocks' encoded versions, f_n and g_n , to the n th server over a classical noiseless channel analogous to the classical PDMM protocol. Once the n th server receives the encoded versions, it can apply classical operations and encode over its own quantum state, ρ_n^{init} , to produce another quantum state ρ_n , i.e.,

$$\rho_n = \text{Enc}(\rho_n^{\text{init}}, f_n, g_n). \quad (6)$$

Then, each server transmits ρ_n to the user over a noise-free quantum channel. Upon receiving all quantum subsystems, the user applies an appropriate measurement to recover the classical output, Y , from the quantum states to decode the required multiplication, AB , i.e.,

$$H(AB|Y) = 0. \quad (7)$$

Remark 2 *Our encoding and decoding approaches follow the N -sum-box framework described in [19].*

The efficiency, i.e., rate, for any scheme satisfying (3) and (7) is defined as the ratio between the number of computations and the number of downloaded symbols, i.e.,

$$R(K, L, T) = \frac{mKL}{N}, \quad (8)$$

where m is an integer representing the number of instances for AB that can be computed. This is a generalization for the rate formulation; however, in the classical setting, we have $m = 1$ and the classical rate is defined as $R_C(K, L, T)$. In the quantum setting, $m = 2$, which will become clear in later sections. The quantum rate is denoted as $R_Q(K, L, T)$. It is important to note that the number of servers in the quantum setting can potentially be different than the number of servers in the classical setting.

III. MAIN RESULTS

We first note that for all the schemes, we define the vector $\alpha = (\alpha_1, \alpha_2)$, similarly for $\beta = (\beta_1, \beta_2)$.

Theorem 1 Let $K \geq L \geq T$, with $2mL \geq \delta L + (m-2)(T-1) + (m-1)L^2 + 6$. Then, there exists a quantum code that is based on a polynomial code that satisfies the PDMM requirements, achieving a rate equal to

$$R_Q = \frac{2KL}{K(L+3) + (m+2)(T-1) + L^2 - 2L}, \quad (9)$$

where $K = m(L-1) + \delta$ and $0 \leq \delta < L-1$. The following are the achievable exponents

$$\alpha_1 = [0 : K-1], \quad (10)$$

$$\beta_1 = [0 : T-1], \quad (11)$$

$$\alpha'_1 = [2K + T + L^2 - 2L - 1, 2K + T + L^2 - 2L, \dots, 2K + T + L^2 - L - 3], \quad (12)$$

$$\alpha'_2 = [3K + 2T + 2L^2 - 2L - 3, 3K + 2T + 2L^2 - 2L - 2, \dots, 3K + 2T + 2L^2 - 2L - 5], \quad (13)$$

$\vdots$

$$\alpha'_m = [(m+1)K + mT + mL^2 - (m+1)L - 2m + 1, (m+1)K + mT + mL^2 - (m+1)L - 2m + 2, \dots, (m+1)K + mT + mL^2 - mL - 2m - 1], \quad (14)$$

$$\alpha'' = [\mu, \mu + 1, \dots, \mu + \delta - 1] \quad (15)$$

$$\beta_2 = [K + T - 2, K + T + L - 3, \dots, K + T + L^2 - 2L - 1], \quad (16)$$

where $\mu = (m+2)K + (m+1)T + (m+1)L^2 - (m+2)L - 2m - 1$, and $\alpha_2 = [\alpha'_1, \dots, \alpha'_m, \alpha'']$, with each α'_i , $i \in [m]$ is of length $L-1$, and α'' is of length δ . The encoding polynomials are given by

$$f(x) = \sum_{i=1}^K R_i x^{\alpha(i)} + \sum_{j=1}^K A_j x^{\alpha(K+j)}, \quad (17)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{j=1}^L B_j x^{\beta(T+j)}. \quad (18)$$

Corollary 1 If $K = L > T$ with $L + T \geq 7$, then there exists a quantum code that is based on a polynomial code that satisfies the PDMM requirements, achieving a rate

$$R_Q = \frac{2L^2}{2L^2 + L + 3T - 3}, \quad (19)$$

with the following exponents

$$\alpha_1 = [0 : L-1], \quad \beta_1 = [0 : T-1], \quad (20)$$

$$\alpha_2 = \underbrace{[L^2 + T - 1, L^2 + T, \dots, L^2 + L + T - 3]}_{L-1},$$

$$2L^2 + 2T - 3], \quad (21)$$

$$\beta_2 = [T + L - 2, T + 2L - 3, \dots, L^2 - L + T - 1], \quad (22)$$

and the encoding polynomials are

$$f(x) = \sum_{i=1}^L R_i x^{\alpha(i)} + \sum_{j=1}^L A_j x^{\alpha(T+j)}, \quad (23)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{j=1}^L B_j x^{\beta(T+j)}. \quad (24)$$

Remark 3 Tables I and II show the degree tables for Theorem 1 and Corollary 1. As opposed to the high privacy regime, the information subspace comes in between the interference subspace.

Remark 4 It is important to note that the matrices A and B are encoded over the high exponents values. This is completely contrary to the classical schemes for PDMM and for the quantum schemes for the high privacy regimes [18] as well as to the quantum versions of the classical codes, as in [17].

To understand the ratio between the highest classical rate and our achieved rate in Theorem 1, Figs. 1-4 show numerical results for the $K = L > T$ case, as this case is simpler to understand and analyze. The figures show that although the privacy parameter is low, it is still possible to achieve gains, compared to the state-of-the-art classical schemes, in terms of rate. Since T is low, it is believed that the problem converges to a communication problem rather than a privacy problem. This poses a challenge to having any benefit in the case of entangled transmitters only, since the gain in this setting occurs from the ability to drop exactly half of the received symbols in the measurement stage. Although these previous two conditions, low privacy and dropping half the transmitted symbols, are conflicting, we were able to have gains up to 1.5 in some cases, as shown in the figures.

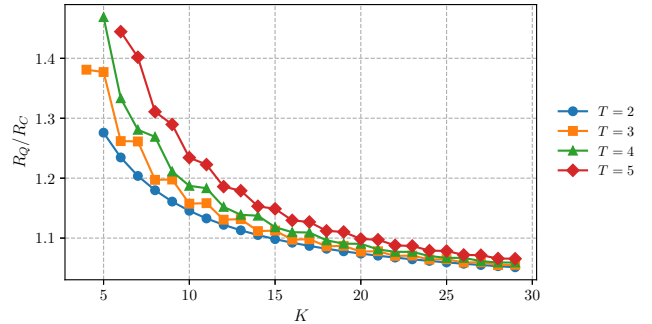


Fig. 1: $K = L$ and $T = [2 : 5]$.

IV. SCHEME FOR $K = L \geq T$

As an illustrative example, we provide the scheme for the case when $K = L > T$ for the result presented in Corollary 1. The general scheme for Theorem 1 follows the same line but it will be cumbersome to present here and not beneficial in terms of building a good intuition.

$\alpha \backslash \beta$	0	1	$\dots$	$T-1$	$K+T-2$	$K+T+L-3$	$\dots$	$K+T+L^2-2L-1$
0	0	$\dots$	$\dots$	$T-1$	$K+T-2$	$\dots$	$\dots$	$K+T+L^2-2L-1$
1	$\vdots$	$\ddots$	$\ddots$	$\vdots$	$\vdots$	$\ddots$	$\ddots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$K-1$	$K-1$	$\dots$	$\dots$	$K+T-2$	$2K+T-3$	$\dots$	$\dots$	$2K+T+L^2-2L-2$
$2K+T+L^2-2L-1$	$2K+T+L^2-2L-1$	$\dots$	$\dots$	$2K+2T+L^2-2L-2$	$3K+2T+L^2-2L-3$	$\dots$	$\dots$	$3K+2T+2L^2-4L-2$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$2K+T+L^2-L-3$	$2K+T+L^2-L-3$	$\dots$	$\dots$	$2K+2T+L^2-L-4$	$3K+2T+L^2-L-5$	$\dots$	$\dots$	$3K+2T+2L^2-3L-4$
$3K+2T+2L^2-2L-3$	$3K+2T+2L^2-2L-3$	$\dots$	$\dots$	$3K+3T+2L^2-2L-4$	$4K+3T+2L^2-2L-5$	$\dots$	$\dots$	$4K+3T+3L^2-4L-4$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$3K+2T+2L^2-L-5$	$3K+2T+2L^2-L-5$	$\dots$	$\dots$	$3K+3T+2L^2-2L-6$	$4K+3T+2L^2-2L-7$	$\dots$	$\dots$	$4K+3T+3L^2-3L-6$
$\vdots$	$\dots$	$\dots$	$\dots$	$\vdots$	$\dots$	$\dots$	$\dots$	$\dots$
$(m+1)K+mT+mL^2-(m+1)L-2m+1$	$(m+1)K+mT+mL^2-(m+1)L-2m+1$	$\dots$	$\dots$	$(m+1)K+(m+1)T+mL^2-(m+1)L-2m$	$(m+2)K+(m+1)T+mL^2-(m+1)L-2m-1$	$\dots$	$\dots$	$(m+2)K+(m+1)T+(m+1)L^2-(m+3)L-2m$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$(m+1)K+mT+mL^2-mL-2m-1$	$(m+1)K+mT+mL^2-mL-2m-1$	$\dots$	$\dots$	$(m+1)K+(m+1)T+mL^2-mL-2m$	$(m+2)K+(m+1)T+mL^2-mL-2m-3$	$\dots$	$\dots$	$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m-2$
$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m-1$	$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m-1$	$\dots$	$\dots$	$(m+2)K+(m+2)T+(m+1)L^2-(m+2)L-2m-2$	$(m+3)K+(m+2)T+(m+1)L^2-(m+2)L-2m-3$	$\dots$	$\dots$	$(m+3)K+(m+2)T+(m+2)L^2-(m+4)L-2m-1$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m+\delta-2$	$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m+\delta-2$	$\dots$	$\dots$	$(m+2)K+(m+2)T+(m+1)L^2-(m+2)L-2m+\delta-3$	$(m+3)K+(m+2)T+(m+1)L^2-(m+2)L-2m+\delta-4$	$\dots$	$\dots$	$(m+3)K+(m+2)T+(m+2)L^2-(m+4)L-2m+\delta-3$

TABLE I: Degree table for $K \geq L > T$.

$\alpha \backslash \beta$	0	1	$\dots$	$T-1$	$T+L-2$	$T+2L-3$	$\dots$	$L^2-L+T-1$
0	0	$\dots$	$\dots$	$T-1$	$T+L-2$	$\dots$	$\dots$	$L^2-L+T-1$
1	$\vdots$	$\ddots$	$\ddots$	$\vdots$	$\vdots$	$\ddots$	$\ddots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$L-1$	$L-1$	$\dots$	$\dots$	$T+L-2$	$T+2L-3$	$\dots$	$\dots$	L^2+T-2
L^2+T-1	L^2+T-1	$\dots$	$\dots$	L^2+2T-2	$L^2+L+2T-3$	$\dots$	$\dots$	$2L^2-L+2T-2$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$L^2+L+T-3$	$L^2+L+T-3$	$\dots$	$\dots$	$L^2+L+2T-4$	$L^2+2L+2T-5$	$\dots$	$\dots$	$2L^2+2T-4$
$2L^2+2T-3$	$2L^2+2T-3$	$\dots$	$\dots$	$2L^2+3T-4$	$2L^2+3T+L-5$	$2L^2+3T+2L-6$	$\dots$	$3L^2-L+3T-4$

TABLE II: Degree table for $K = L > T$.

The n th server receives two instances of the polynomials in (23) and (24) evaluated at a_n and multiplies them to get

$$F^{[m]}(a_n) = \sum_{(i,j) \in \mathcal{Q}} I_{i,j}^{[m]} a_n^{(\alpha \oplus \beta)_{i,j}}$$

$$+ \sum_{(k,\ell) \in \bar{\mathcal{Q}}} A_k^{[m]} B_\ell^{[m]} a_n^{(\alpha \oplus \beta)_{k,\ell}}, \quad m \in [2], \quad (25)$$

where $\mathcal{Q}$ is the interference subspace with $i \in [0 : L-1] \cup \{2L^2+2T-3\}$, and $j \in [0 : T-1] \cup \{T+L-2, \dots, L^2-L+T-1\}$. Now, for each element in $F_n^{[1]} = F^{[1]}(a_n)$, $F_n^{[2]} =$

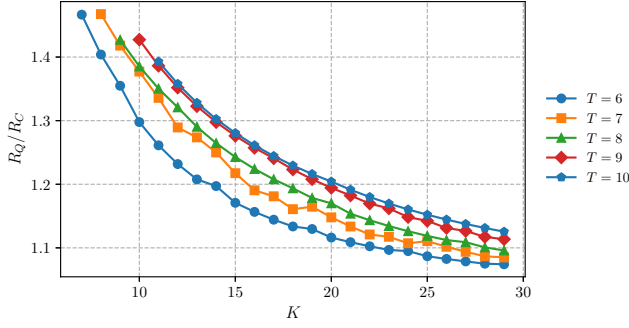


Fig. 2: $K = L$ and $T = [6 : 10]$.

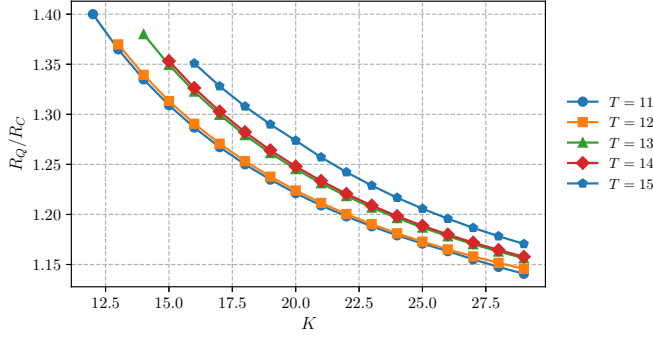


Fig. 3: $K = L$ and $T = [11 : 15]$.

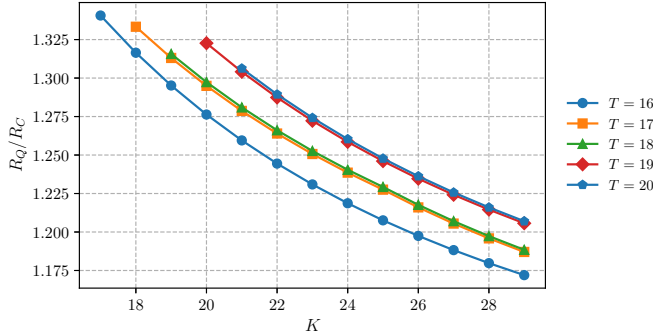


Fig. 4: $K = L$ and $T = [16 : 20]$.

$F^{[2]}(a_n)$, the server applies Pauli operators on each element

$$\begin{aligned} & \mathbf{X}(u_n F_n^{[1]}(i, j)) \mathbf{Z}(v_n F_n^{[2]}(i, j)) \rho_n \\ & \times \mathbf{X}^\dagger(u_n F_n^{[1]}(i, j)) \mathbf{Z}^\dagger(v_n F_n^{[2]}(i, j)). \end{aligned} \quad (26)$$

The transmitted symbols can be written as follows,

$$\begin{aligned} X(i, j) &= [F_{[1:2L^2+L+3T-3]}^{[1]}(i, j), F_{[1:2L^2+L+3T-3]}^{[2]}(i, j)]^t \\ &= \begin{bmatrix} \text{diag}(u)G' & 0_{2L^2+L+3T-3} \\ 0_{2L^2+L+3T-3} & \text{diag}(v)G' \end{bmatrix} \\ & \times [\bar{I}^{[1]}(i, j), (\overline{AB})^{[1]}(i, j), \underline{I}^{[1]}(i, j), (\underline{AB})^{[1]}(i, j), \end{aligned} \quad (27)$$

$$\bar{I}^{[2]}(i, j), (\overline{AB})^{[2]}(i, j), \underline{I}^{[2]}(i, j), (\underline{AB})^{[2]}(i, j)]^t, \quad (28)$$

where

$$G' = \begin{bmatrix} 1 & a_1 & a_1^2 & \dots & a_1^c & a_1^d & a_1^{d+L-1} & \dots & a_1^e \\ 1 & a_2 & a_2^2 & \dots & a_2^c & a_2^d & a_2^{d+L-1} & \dots & a_2^e \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_N & a_N^2 & \dots & a_N^c & a_N^d & a_N^{d+L-1} & \dots & a_N^e \end{bmatrix}, \quad (29)$$

where $c = 2L^2 + 3T - 4$, $d = 2L^2 + 3T + L - 5$, $e = 3L^2 - L + 3T - 4$ and $N = 2L^2 + L + 3T - 3$. In addition, using the notation in (25) and the mapping $(i, j) \rightarrow i + j$ in the subscript $\bar{I} = I_{[0:L^2+L+2T-4]}$, $\overline{AB} = [A_1 B_1, A_1 B_2, \dots, A_1 B_L, \dots, A_{L-1} B_1, \dots, A_{L-1} B_L]$, $\underline{I} = I_{[2L^2+2T-3:2L^2+3T-4]}$ and $\underline{AB} = [A_L B_1, A_L B_2, \dots, A_L B_L]$. Once the user receives the quantum dits and applies the measurements, the received classical dits can be written as

$$Y(i, j) = MX(i, j) \quad (30)$$

$$\begin{aligned} &= \underbrace{\left[I_{\lfloor \frac{N}{2} \rfloor - 1}^{[1]}(i, j), \dots, I_{L^2+L+2T-4}^{[1]}(i, j), \right.}_{\text{Remaining interference subspace 1}} \\ & \quad (\overline{AB})^{[1]}(i, j), \underline{I}^{[1]}(i, j), (\underline{AB})^{[1]}(i, j), \\ & \quad \left. I_{\lceil \frac{N}{2} \rceil - 1}^{[2]}(i, j), \dots, I_{L^2+L+2T-4}^{[2]}(i, j), \right.}_{\text{Remaining interference subspace 2}} \\ & \quad \left. (\overline{AB})^{[2]}(i, j), \underline{I}^{[2]}(i, j), (\underline{AB})^{[2]}(i, j) \right]^t, \end{aligned} \quad (31)$$

where $M = \begin{bmatrix} 0 & I_N \end{bmatrix} \begin{bmatrix} G & H \end{bmatrix}^{-1}$, with

$$G = D \begin{bmatrix} G'_1 & 0 \\ 0 & G'_2 \end{bmatrix}, \quad (32)$$

with $G'_1 = G'(:, 1 : \lfloor \frac{N}{2} \rfloor - 1)$, $G'_2 = G'(:, 1 : \lceil \frac{N}{2} \rceil - 1)$ and

$$H = D \begin{bmatrix} H_1 & 0 \\ 0 & H_2 \end{bmatrix}, \quad (33)$$

with $H_1 = G'(:, \lfloor \frac{N}{2} \rfloor : \text{end})$, $H_2 = G'(:, \lceil \frac{N}{2} \rceil : \text{end})$, and

$$D = \begin{bmatrix} \text{diag}(u) & 0 \\ 0 & \text{diag}(v) \end{bmatrix}. \quad (34)$$

As shown in (30), the remaining interference subspace has lower dimensionality compared to the one in (27) for $\bar{I}^{[1]}(i, j)$ and $\bar{I}^{[2]}(i, j)$. The condition that $L+T \geq 7$ ensures that the information subspace remains unaffected with the measurement decoding over-the-air, as follows, since $\frac{L+T}{2} \geq 3.5$, we have

$$\Rightarrow L^2 + L + 2T - 4 \geq L^2 + \frac{L}{2} + 1.5T - 0.5 \quad (35)$$

$$\Rightarrow L^2 + L + 2T - 4 \geq \frac{2L^2 + L + 3T - 3}{2} + 1 \quad (36)$$

$$\Rightarrow L^2 + L + 2T - 4 \geq \left\lceil \frac{2L^2 + L + 3T - 3}{2} \right\rceil. \quad (37)$$

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